

Modulus, argument form

Basic skills

Q1. a) $z = 8 + 6i$

$$|z| = \sqrt{8^2 + 6^2} = \sqrt{100} = 10$$

b) $z = 4 - i$

$$|z| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

c) $z = 4i$

$$|z| = \sqrt{0^2 + 4^2} = \sqrt{16} = 4$$

Q2. a) $z = 4 + 4i$

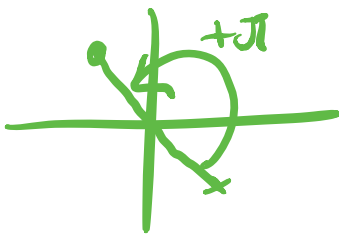
$$\text{Arg}(z) = \tan^{-1}\left(\frac{4}{4}\right) = \frac{\pi}{4}$$

b) $z = 2i - 1$

$$\text{Arg}(z) = \tan^{-1}\left(\frac{2}{-1}\right) = -1.107$$

but $x < 0, y > 0 \downarrow +\pi$

so $\text{Arg}(z) = \pi - 1.107$

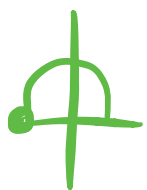


c) $z = -4i$

$$\text{Arg}(z) = -\frac{\pi}{2}$$

← as $x = 0$ can't use formula





$$2) z = -5$$

$$\text{Arg}(z) = \pi$$

← can use $\tan^{-1}(0) = 0$
then $+\pi$

Competency skills

Q3. By triangle $x = |z| \cos(\text{Arg } z)$
 $y = |z| \sin(\text{Arg } z)$

so $z = |z| \cos(\text{Arg } z) + |z| \sin(\text{Arg } z) i$
 $= |z| (\cos(\text{Arg } z) + i \sin(\text{Arg } z))$

Q4. a) $z = 4 + 4i$

$$|z| = \sqrt{4^2 + 4^2} = 4\sqrt{2}$$

$$\text{Arg } z = \tan^{-1}\left(\frac{4}{4}\right) = \frac{\pi}{4}$$

so $z = 4\sqrt{2} \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right)$

b) $z = 3 - 4i$

$$|z| = \sqrt{3^2 + 4^2} = 5$$

$$\text{Arg } z = \tan^{-1}\left(-\frac{4}{3}\right) = -0.927$$

$$\text{SO } z = 5 (\cos(-0.927) + i \sin(-0.927))$$

$$c) \quad z = i - \sqrt{3}$$

$$|z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$$\text{Arg } z = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = -\frac{\pi}{6} \quad \begin{array}{l} \text{but } x < 0 \\ y > 0 \text{ so} \end{array}$$

$$\text{Arg } z = \frac{5\pi}{6} \quad \leftarrow +\pi$$

$$z = 2 \left(\cos\left(\frac{5\pi}{6}\right) + i \sin\left(\frac{5\pi}{6}\right) \right)$$

$$d) \quad z = -i - \sqrt{3}$$

$$|z| = \sqrt{1^2 + (\sqrt{3})^2} = 2$$

$$\text{Arg } z = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$\text{Arg } z = -\frac{5\pi}{6}$$

$$\swarrow \quad \begin{array}{l} x < 0, \\ y < 0 \end{array}$$

$$z = 2 \left(\cos\left(-\frac{5\pi}{6}\right) + i \sin\left(-\frac{5\pi}{6}\right) \right)$$

Q5. $z = \frac{1}{2} - 2i$

a) $|z| = \sqrt{\frac{1}{4} + 4} = \sqrt{\frac{17}{4}} = \frac{\sqrt{17}}{2}$

$\text{Arg } z = \tan^{-1}\left(\frac{-2}{\frac{1}{2}}\right) = \tan^{-1}(-4) = -1.326$

b) $2z = 1 - 4i$

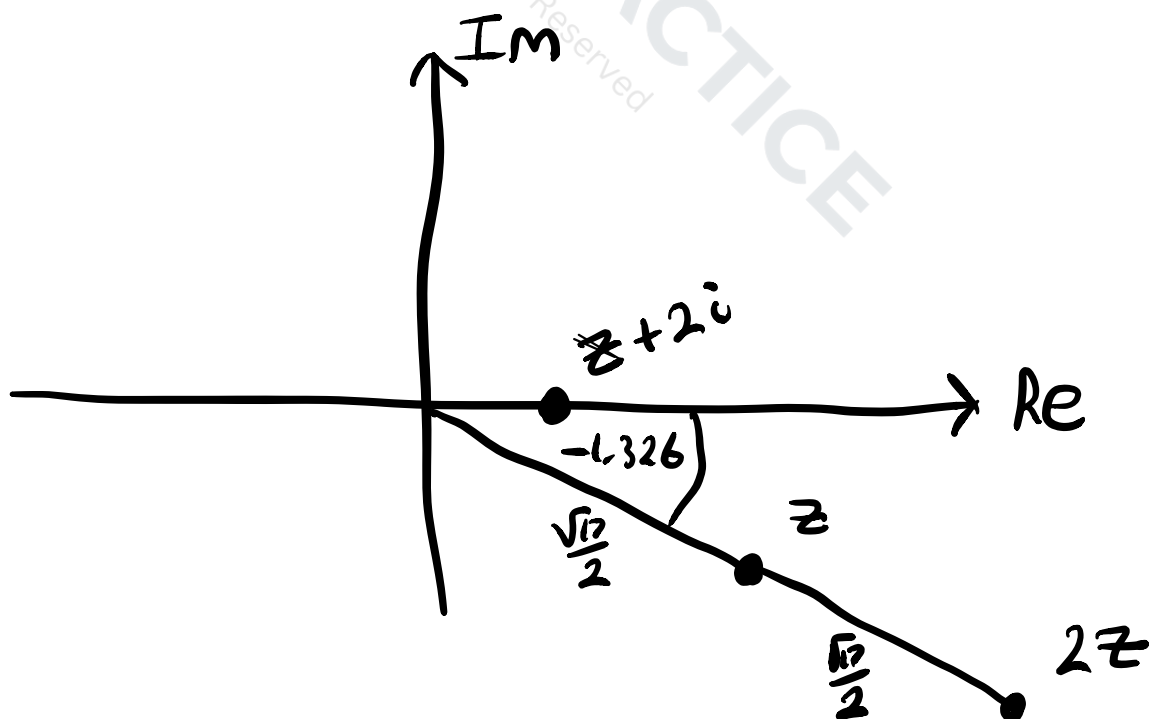
or $\sqrt{1^2 + 4^2}$

$|2z| = 2|z| = \sqrt{17}$

$\text{Arg}(2z) = \text{Arg } z = -1.326$ or $\tan^{-1}\left(\frac{-4}{1}\right)$

c) $z + 2i = \frac{1}{2}$

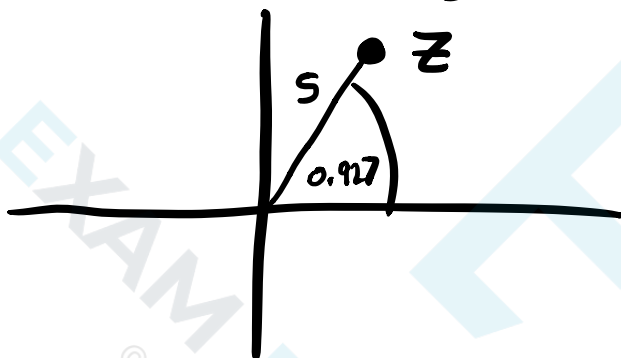
$|z + 2i| = \frac{1}{2}$ $\text{Arg}(z + 2i) = 0$



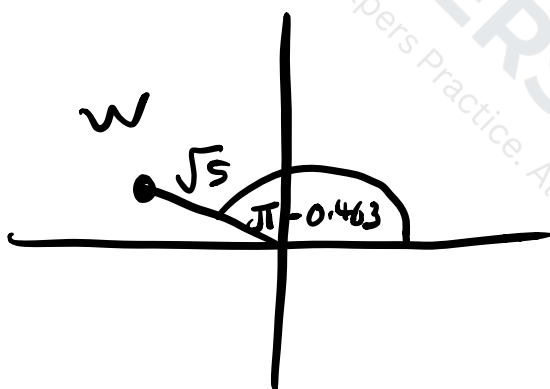
Advanced skills

Q 6. $z = 3 + 4i$, $w = i - 2$

a) $|z| = 5$ $\text{Arg } z = \tan^{-1}\left(\frac{4}{3}\right) = 0.927$



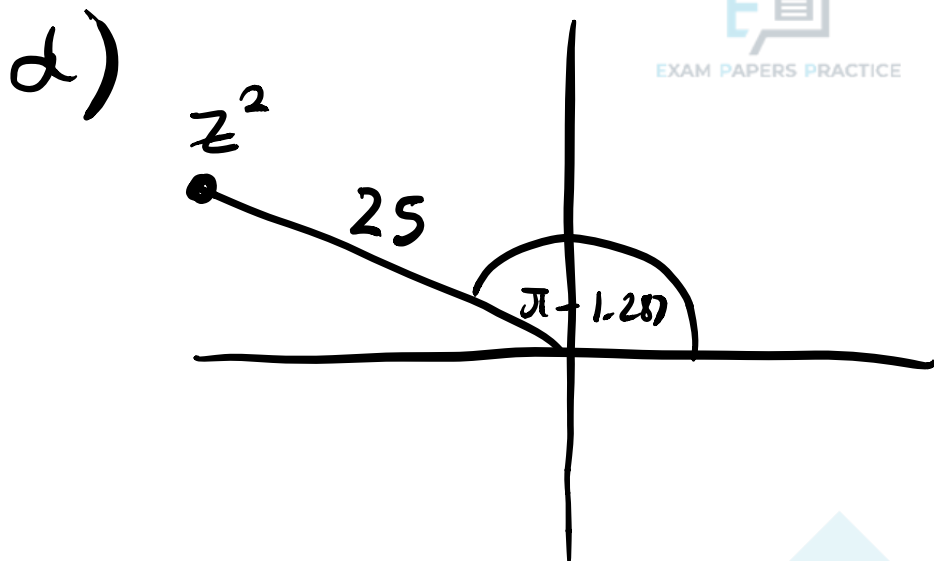
b) $|w| = \sqrt{5}$ $\text{Arg } w = \tan^{-1}\left(-\frac{1}{2}\right) = \pi - 0.463$



c) $z^2 = (3 + 4i)^2 = 9 - 16 + 24i$
 $= -7 + 24i$

$|z^2| = \sqrt{7^2 + 24^2} = \sqrt{625} = 25$

$\text{Arg}(z^2) = \tan^{-1}\left(\frac{24}{-7}\right) = -1.287$
 $+ \pi$



e)

$$zw = (3+4i)(i-2)$$

$$= 3i - 8i - 6 - 4$$

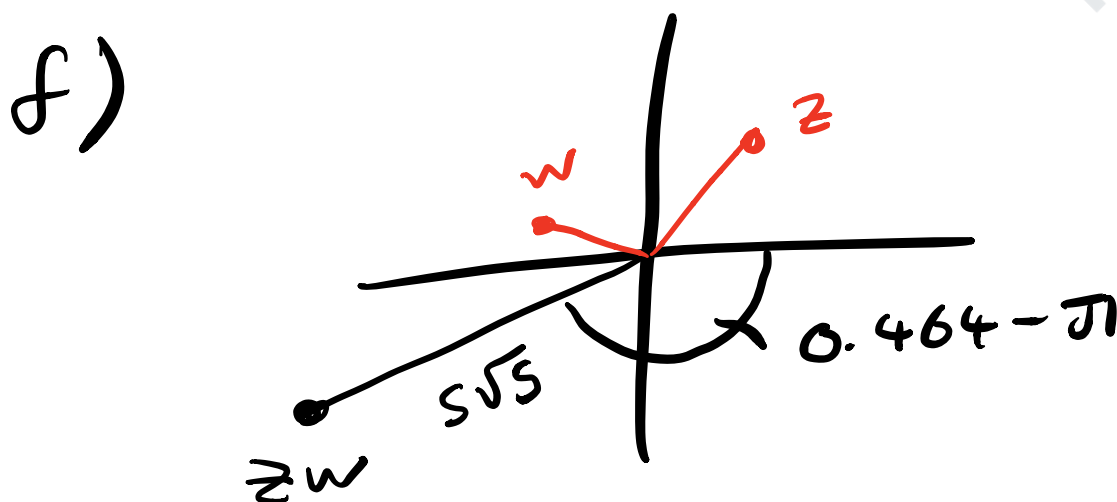
$$= -10 - 5i$$

$$|zw| = \sqrt{10^2 + 5^2} = \sqrt{125} = 5\sqrt{5}$$

$$\text{Arg}(zw) = \tan^{-1}\left(\frac{-5}{-10}\right) = 0.464 - \pi$$

$$zw = 5\sqrt{5} \left(\cos(0.464 - \pi) + i \sin(0.464 - \pi) \right)$$

π
 $x < 0,$
 $y < 0$



g) True as

$$5\sqrt{5} = 5 \times \sqrt{5} \text{ and}$$

$$\begin{aligned} 0.927 + \pi - 0.463 &= \pi + 0.464 \\ &= 0.464 - \pi \end{aligned}$$

*-2\pi
+0
get
in same
form*